

MainActivity.java

```
package com.example.crud;

import android.app.AlertDialog;
import android.database.Cursor;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import android.widget.Toast;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    DatabaseHelper myDb;

    EditText editId, editName, editAge;

    Button btnAdd, btnView, btnUpdate, btnDelete;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);
        myDb = new DatabaseHelper(this);
        editId = findViewById(R.id.editId);
        editName = findViewById(R.id.editName);
        editAge = findViewById(R.id.editAge);
        btnAdd = findViewById(R.id.btnAdd);
        btnView = findViewById(R.id.btnView);
        btnUpdate = findViewById(R.id.btnUpdate);
        btnDelete = findViewById(R.id.btnDelete);

        addData();
        viewAll();
        updateData();
    }
}
```

```

deleteData();

ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {
    Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());
    v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);
    return insets;
});
}

public void addData() {
    btnAdd.setOnClickListener(v -> {
        boolean isInserted = myDb.insertData( editName.getText().toString(),
            editAge.getText().toString());
        Toast.makeText(this, isInserted ? "Data Inserted" : "Data not Inserted",
            Toast.LENGTH_SHORT).show();
    });
}

public void viewAll() {
    btnView.setOnClickListener(v -> {
        Cursor res = myDb.getAllData();
        if (res.getCount() == 0) {
            showMessage("Error", "Nothing found");
            return;
        }

        StringBuilder buffer = new StringBuilder();
        while (res.moveToNext()) {
            buffer.append("ID: ").append(res.getString(0)).append("\n");
            buffer.append("Name: ").append(res.getString(1)).append("\n");
            buffer.append("Age: ").append(res.getString(2)).append("\n\n");
        }
        showMessage("Data", buffer.toString());
    });
}

public void updateData() {
    btnUpdate.setOnClickListener(v -> {

```

```

        boolean isUpdated = myDb.updateData( editId.getText().toString(), editName.getText().toString(),
editAge.getText().toString());

        Toast.makeText(this,isUpdated ? "Data Updated" : "Data not Updated",
            Toast.LENGTH_SHORT).show();

    });

}

public void deleteData() {

    btnDelete.setOnClickListener(v -> {

        int deletedRows = myDb.deleteData(editId.getText().toString());

        Toast.makeText(this, deletedRows > 0 ? "Data Deleted" : "Data not Deleted",
            Toast.LENGTH_SHORT).show();

    });

}

public void showMessage(String title, String message) {

    AlertDialog.Builder builder = new AlertDialog.Builder(this);

    builder.setCancelable(true);

    builder.setTitle(title);

    builder.setMessage(message);

    builder.show();

}

}

```

DatabaseHelper.java

```

package com.example.crud;

import android.content.ContentValues;
import android.content.Context;
import android.database.Cursor;
import android.database.sqlite.SQLiteDatabase;
import android.database.sqlite.SQLiteOpenHelper;

public class DatabaseHelper extends SQLiteOpenHelper {

    private static final String DATABASE_NAME = "student.db";

    private static final String TABLE_NAME = "students";

    private static final String COL_1 = "ID";

    private static final String COL_2 = "NAME";

    private static final String COL_3 = "AGE";

```

```

public DatabaseHelper(Context context) {
    super(context, DATABASE_NAME, null, 1);
}

@Override
public void onCreate(SQLiteDatabase db) {
    db.execSQL("CREATE TABLE " + TABLE_NAME +
        "(ID INTEGER PRIMARY KEY AUTOINCREMENT, NAME TEXT, AGE TEXT)");
}

@Override
public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {
    db.execSQL("DROP TABLE IF EXISTS " + TABLE_NAME);
    onCreate(db);
}

// INSERT
public boolean insertData(String name, String age) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues contentValues = new ContentValues();
    contentValues.put(COL_2, name);
    contentValues.put(COL_3, age);
    long result = db.insert(TABLE_NAME, null, contentValues);
    return result != -1;
}

// READ
public Cursor getAllData() {
    SQLiteDatabase db = this.getWritableDatabase();
    return db.rawQuery("SELECT * FROM " + TABLE_NAME, null);
}

// UPDATE
public boolean updateData(String id, String name, String age) {
    SQLiteDatabase db = this.getWritableDatabase();
    ContentValues contentValues = new ContentValues();
    contentValues.put(COL_1, id);
    contentValues.put(COL_2, name);
    contentValues.put(COL_3, age);

```

```

        db.update(TABLE_NAME, contentValues, "ID = ?", new String[]{id});
        return true;
    }
    // DELETE
    public Integer deleteData(String id) {
        SQLiteDatabase db = this.getWritableDatabase();
        return db.delete(TABLE_NAME, "ID = ?", new String[]{id});
    }
}

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    tools:context=".MainActivity" >
    <EditText
        android:id="@+id/editId"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:hint="id"
        android:inputType="text" />
    <EditText
        android:id="@+id/editName"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:ems="10"
        android:hint="name"
        android:inputType="text" />
    <EditText

```

```
    android:id="@+id/editAge"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:ems="10"
    android:hint="age"
    android:inputType="text" />
<Button
    android:id="@+id/btnAdd"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Add" />
<Button
    android:id="@+id/btnView"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="View" />
<Button
    android:id="@+id/btnUpdate"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Update" />
<Button
    android:id="@+id/btnDelete"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Delete" />
</LinearLayout>
```